

(AR) Auto regressive model



$$y_t = \pi_1 y_{t-1} + \pi_2 y_{t-2} + \dots + z_t$$

$$(y_t - \pi_1 y_{t-1} - \pi_2 y_{t-2} - \dots) = z_t$$

By using Backward operators

$$(y_t - \pi_1 B y_{t-1} - \pi_2 B^2 y_t - \dots) = z_t$$

$$(1 - \pi_1 B - \pi_2 B^2 - \dots) = z_t$$

$$\pi(B) y_t = z_t \quad \text{--- (i)}$$

$$\pi(B) = 1 - \sum_{i=1}^{\infty} \pi_i B^i$$

eq (i) can be written as:

$$y_t = \pi^{-1}(B) z_t$$

$$\text{where } \pi(B) \Rightarrow 1 + \sum_{i=1}^{\infty} |\pi_i| < \infty$$

$$\sum_{i=0}^{\infty} |\pi_i| < \infty$$

In the autoregressive representation of a process if only a finite number of ' π ' weights are non-zero i.e

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$$\pi_1 = \phi_1, \pi_2 = \phi_2, \dots, \pi_p = \phi_p \quad \text{and}$$

$\pi_k = 0$ for $k > p$, then the resulting

process is said to be an autoregressive process (model) of order p , which is denoted as $AR(p)$. It is given by:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + z_t$$

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$p+1$ parameters are estimated.
If μ is included then $p+2$ parameters are estimated.

✱ ————— ✱

$\phi_1, \phi_2, \dots, \phi_p$ are the finite set of weight parameters.

AR(1) model:

$$y_t = \phi_1 y_{t-1}$$

AR(2) model:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + z_t$$

AR(3) Model:

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$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + z_t$$

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AR(p) Model:

The autoregressive process of order (p) is defined as,

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + Z_t$$

$$Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} - \dots - \phi_p Y_{t-p} = Z_t$$

By using backward operator

$$Y_t - \phi_1 B Y_t - \phi_2 B^2 Y_t - \dots - \phi_p B^p Y_t = Z_t$$

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p) Y_t = Z_t$$

Polynomial $\rightarrow \phi_p(B) Y_t = Z_t$ — (i)

$$\phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$$

$$\phi_p(B) = 1 - \sum_{i=1}^p \phi_i B^i$$

From eq (i)

$$Y_t = [\phi_p(B)]^{-1} Z_t$$

By putting value

$$= \left[1 - (\phi_1 B + \phi_2 B^2 + \dots + \phi_p B^p) \right]^{-1} Z_t \quad (83)$$

By using $(1-x)^{-1}$ series.

$$\therefore (1-x)^{-1} = 1 + x + x^2 + \dots$$

$$Y_t = \left[1 + (\phi_1 B + \phi_2 B^2 + \dots + \phi_p B^p) + (\phi_1 B + \phi_2 B^2 + \dots + \phi_p B^p)^2 + (\phi_1 B + \phi_2 B^2 + \dots + \phi_p B^p)^3 + \dots \right] Z_t$$

$$Y_t = \left[1 + \phi_1 B + (\phi_2 + \phi_1^2) B^2 + \dots \right] Z_t$$

$$Y_t = Z_t + \phi_1 B Z_t + (\phi_2 + \phi_1^2) B^2 Z_t + \dots$$

$$Y_t = Z_t + \phi_1 Z_{t-1} + (\phi_2 + \phi_1^2) Z_{t-2} + \dots$$

$$Y_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \dots$$

$$Y_t = \sum_{i=0}^{\infty} \theta_i Z_{t-i}$$

From notes
at page # 56

i) Mean of AR(p)

$$Y_t = \sum_{i=0}^{\infty} \theta_i Z_{t-i}$$

Apply Expectation.

$$E(Y_t) = \sum_{i=0}^{\infty} \alpha_i E(Z_{t-i}) \quad (84)$$

$$= 0$$

$$\because E(Z_{t-i}) = 0$$

ii) Variance

$$\text{Variance} = E(Y_t)^2 - [E(Y_t)]^2$$

$$= E(Y_t^2) - 0$$

$$= E(Y_t^2)$$

$$= E\left(\sum_{i=0}^{\infty} \alpha_i Z_{t-i}\right)^2$$

$$= E\left[\sum_{i=0}^{\infty} \alpha_i^2 Z_{t-i}^2 + \sum_{i \neq j} \alpha_i \alpha_j Z_{t-i} Z_{t-j}\right]$$

$$= \sum_{i=0}^{\infty} \alpha_i^2 E(Z_{t-i}^2) + 0$$

$$\text{Variance} = \sigma_z^2 \sum_{i=0}^{\infty} \alpha_i^2$$

Variance will be finite if $\sum_{i=0}^{\infty} \alpha_i^2$ is finite. Thus we need some conditions on parameters of AR process for its stationary.

iii) AutoCovariance

The autoCovariance of AR(p) is defined as

$$\gamma_k = \text{Cov}(Y_t, Y_{t-k})$$

$$= E(Y_t Y_{t-k}) - E(Y_t) E(Y_{t-k})$$

$$= E(Y_t Y_{t-k}) - 0 \quad \because E(Y_t) = 0 = E(Y_{t-k})$$

$$\gamma_k = E(Y_t Y_{t-k})$$

$$\text{As } Y_t = \sum_{i=0}^{\infty} \alpha_i Z_{t-i}$$

$$Y_{t-k} = \sum_{j=0}^{\infty} \alpha_j Z_{t-k-j}$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \alpha_i Z_{t-i} \sum_{j=0}^{\infty} \alpha_j Z_{t-k-j} \right]$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \sum_{\substack{j=0 \\ j=i-k}}^{\infty} \alpha_i \alpha_j Z_{t-i} Z_{t-j-k} + \sum_{i=0}^{\infty} \sum_{\substack{j=0 \\ j \neq i-k}}^{\infty} \alpha_i \alpha_j Z_{t-i} Z_{t-j-k} \right]$$

$$\gamma_k = E \left[\sum_{i=k}^{\infty} \alpha_i \alpha_{i-k} Z_{t-i}^2 + \text{C.P term} \right]$$

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$$\gamma_k = \sum_{i=k}^{\infty} O_i O_{i-k} E(Z_{t-i}^2)$$

$$\gamma_k = \sigma_z^2 \sum_{i=k}^{\infty} O_i O_{i-k} \quad \text{for } k=1, 2, \dots$$

$$\gamma_k = \sigma_z^2 \sum_{i=k}^{\infty} O_i O_{i+k}$$

For Backward
or forward
Auto Covariance is
Same

If we put $k=0$ it
becomes γ_0

iv) Autocorrelation

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

$$= \frac{\sigma_z^2 \sum_{i=k}^{\infty} O_i O_{i+k}}{\sigma_z^2 \sum_{i=0}^{\infty} O_i^2}$$

$$\rho_k = \frac{\sum_{i=k}^{\infty} O_i O_{i+k}}{\sum_{i=0}^{\infty} O_i^2}, \quad k=1, 2, 3, \dots$$

✓

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Stationary Condition of AR process

From
notes
at Page
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Consider an AR(P) process.

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + Z_t$$

$$Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} - \dots - \phi_p Y_{t-p} = Z_t$$

$$Y_t - \phi_1 B Y_t - \phi_2 B^2 Y_t - \dots - \phi_p B^p Y_t = Z_t$$

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p) Y_t = Z_t \quad (i)$$

$$\phi_p(B) Y_t = Z_t$$

The Above process is said to be stationary if all the roots of equation

$$\phi_p(B) = 0$$

lies outside the unit circle.

$$|B_i| > 1 \quad \text{for } i = 1, 2, 3, \dots, p$$

Alternatively, we can state the stationary conditions by using the characteristic equation of AR(P) process.

$$x^p - \phi_1 x^{p-1} - \phi_2 x^{p-2} - \dots - \phi_p = 0 \quad (88)$$

The process will be stationary if all the roots of characteristic equation lie inside the unit circle,

i.e

$$|x_i| < 1 \quad \text{for } i=1, 2, \dots, p$$

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See example from notes at pg # 63 or 77

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Yule-Walker Equations.

Notes pg 66

Q. Derive Yule-Walker equations for an AR(p) process.

Consider a zero mean AR(p) process.

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + Z_t$$

multiply both sides by Y_{t-k}

$$Y_t Y_{t-k} = \phi_1 (Y_{t-1} Y_{t-k}) + \phi_2 (Y_{t-2} Y_{t-k}) + \dots + \phi_p (Y_{t-p} Y_{t-k}) + (Z_t Y_{t-k})$$

Apply Expectation.

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$$E[Y_t Y_{t-k}] = \phi_1 E(Y_{t-1} Y_{t-k}) + \phi_2 E(Y_{t-2} Y_{t-k}) + \dots \\ \dots + \phi_p E(Y_{t-p} Y_{t-k}) + E(Z_t Y_{t-k})$$

For $k > 0$

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \dots + \phi_p \gamma_{k-p} + 0$$

$$\therefore E(Z_t Y_{t-k}) = 0$$

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \dots + \phi_p \gamma_{k-p}$$

Divide both sides by γ_0

$$\frac{\gamma_k}{\gamma_0} = \frac{\phi_1 \gamma_{k-1}}{\gamma_0} + \frac{\phi_2 \gamma_{k-2}}{\gamma_0} + \dots + \frac{\phi_p \gamma_{k-p}}{\gamma_0}$$

we get

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \dots + \phi_p \rho_{k-p} \quad \text{for } k > 0$$

— (i)

Put $k = 1, 2, 3, \dots, p$ in eq (i)

$$\underline{k=1} \quad \rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1} + \dots + \phi_p \rho_{1-p}$$

$$\underline{k=2} \quad \rho_2 = \phi_1 \rho_1 + \phi_2 \rho_0 + \dots + \phi_p \rho_{2-p}$$

$$\vdots$$

$$\underline{k=p} \quad \rho_p = \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \dots + \phi_p \rho_0$$

As we know that $l_0 = 1, l_{-k} = l_k$

$$l_1 = \phi_1 + \phi_2 l_1 + \dots + \phi_p l_{p-1}$$

$$l_2 = \phi_1 l_1 + \phi_2 + \dots + \phi_p l_{p-2}$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

$$l_p = \phi_1 l_{p-1} + \phi_2 l_{p-2} + \dots + \phi_p$$

In matrix form.

$$\begin{bmatrix} l_1 \\ l_2 \\ \vdots \\ l_p \end{bmatrix} = \begin{bmatrix} 1 & \dots & l_{p-1} \\ l_1 & \dots & l_{p-2} \\ \vdots & \ddots & \vdots \\ l_{p-1} & \dots & l_p \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{bmatrix}$$

$$\underline{l} = \underline{L} \underline{\phi}$$

$$\underline{\phi} = \underline{L}^{-1} \underline{l}$$

For sample data

$$\hat{\underline{\phi}} = \underline{R}^{-1} \underline{r}$$

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where

$$\underline{r} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_p \end{bmatrix}, \quad \underline{R} = \begin{bmatrix} 1 & r_1 & \dots & r_{p-1} \\ r_1 & 1 & \dots & r_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p-1} & r_{p-2} & \dots & 1 \end{bmatrix}$$

The general solution of ' ρ_k ' Yule-walker is given as:

$$\rho_k = A_1 \pi_1^{(k)} + A_2 \pi_2^{(k)} + \dots + A_p \pi_p^{(k)}$$

where

$$\rightarrow \sum_{i=1}^p A_i = 1$$

$\rightarrow \pi_1, \pi_2, \dots, \pi_p$ are the roots of the characteristic equation of AR(p) process.

$$x^p - \phi_1 x^{p-1} - \phi_2 x^{p-2} - \dots - \phi_p = 0$$

For stationarity $|\pi_i| < 1$

✱. ————— ✱

Purpose:

- i) To get the Autocorrelation function
- ii) To estimate parameters of AR process

	stationary	Invertibility
MA	✓	?
AR	?	✓

MA process is stationary but we check its invertibility.

From notes see examples at page 66-84

AR(1) process

Page # 50

The autoregressive process of order 1 is defined as:

$$Y_t = \phi Y_{t-1} + Z_t$$

$$Y_t - \phi Y_{t-1} = Z_t$$

$$Y_t - \phi B Y_t = Z_t$$

$$(1 - \phi B) Y_t = Z_t$$

$$Y_t = (1 - \phi B)^{-1} Z_t$$

$$Y_t = (1 + \phi B + \phi^2 B^2 + \phi^3 B^3 + \dots) Z_t$$

$$(1-x)^{-1} = 1+x+x^2+\dots$$

$$Y_t = \left(\sum_{i=0}^{\infty} \phi^i B^i \right) Z_t$$

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$$Y_t = \sum_{i=0}^{\infty} \phi^i B^i Z_t$$

$$Y_t = \sum_{i=0}^{\infty} \phi^i Z_{t-i}$$

i) mean.

$$Y_t = \sum_{i=0}^{\infty} \phi^i Z_{t-i}$$

$$E(Y_t) = \sum_{i=0}^{\infty} \phi^i E(Z_{t-i})$$

$$= 0$$

ii) variance.

$$\text{Var} = E(Y_t^2) - [E(Y_t)]^2$$

$$= E(Y_t^2) - 0$$

$$= E(Y_t^2)$$

$$= E\left(\sum_{i=0}^{\infty} \phi^i Z_{t-i}\right)^2$$

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$$= E \left[\sum_{i=0}^{\infty} \phi^{2i} z_{t-i}^2 + \sum_{i=0}^{\infty} \sum_{\substack{j=0 \\ i \neq j}}^{\infty} \phi^i \phi^j z_{t-i} z_{t-j} \right]$$

$$= \sum_{i=0}^{\infty} \phi^{2i} E(z_{t-i}^2) + 0$$

$$\text{var} = \sigma_z^2 \sum_{i=0}^{\infty} \phi^{2i}$$

$$\begin{aligned} \text{variance} &= \sigma_z^2 (\phi^0 + \phi^2 + \phi^4 + \phi^6 + \dots) \\ &= \sigma_z^2 (1 + \phi^2 + \phi^4 + \phi^6 + \dots) \end{aligned}$$

It is an infinite Geometric series with common ratio $r = \phi^2$. This series will be convergent if $|r| < 1$ i.e. $|\phi| < 1$

$$\text{variance} = \sigma_z^2 \left(\frac{1}{1 - \phi^2} \right)$$

where $|\phi| < 1$

$S_{\infty} = \frac{a}{1-r}$ $a = \text{first term}$ $r = \text{common ratio}$
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iii) AutoCovariance:

The autocovariance function of AR(1) is defined as:

$$\gamma_k = \text{Cov}(Y_t, Y_{t-k})$$

$$= E(Y_t, Y_{t-k}) - E(Y_t) E(Y_{t-k})$$

$$= E(Y_t Y_{t-k}) - 0 \quad \because E(Y_t) = 0 = E(Y_{t-k})$$

$$\gamma_k = E(Y_t Y_{t-k})$$

$$Y_t = \sum_{i=0}^{\infty} \phi^i Z_{t-i}$$

$$Y_{t-k} = \sum_{j=0}^{\infty} \phi^j Z_{t-k-j}$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \phi^i Z_{t-i} \cdot \sum_{j=0}^{\infty} \phi^j Z_{t-k-j} \right]$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \phi^i \phi^j Z_{t-i} Z_{t-k-j} \right]$$

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$$\gamma_k = E \left[\sum_{i=0}^{\infty} \sum_{\substack{j=0 \\ j \neq i-k}}^{\infty} \phi^i \phi^j z_{t-i} z_{t-k-j} + \sum_{i=0}^{\infty} \sum_{\substack{j=0 \\ j \neq i-k}}^{\infty} \phi^i \phi^j z_{t-i} z_{t-k-j} \right]$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \phi^{i+j} z_{t-i} z_{t-k-j} + \text{C.P. term} \right]$$

$$= \sum_{i=k}^{\infty} \phi^{2i-k} E[z_{t-i}^2] + 0$$

$$= \sigma_z^2 \sum_{i=k}^{\infty} \phi^{2i-k}$$

$$\gamma_k = \sigma_z^2 \left(\phi^k + \phi^{k+2} + \phi^{k+4} + \dots \right)$$

which is geometric series.

$$a = \phi^k, \quad r = \frac{\phi^{k+2}}{\phi^k}$$

$$= \frac{\cancel{\phi^k} \phi^2}{\cancel{\phi^k}}$$

$$r = \phi^2$$

So

$$\gamma_k = \sigma_z^2 \left(\frac{\phi^k}{1 - \phi^2} \right), \quad k = 0, 1, 2, \dots$$

where $|\phi| < 1$

iv) Auto Correlation.

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

$$= \frac{\cancel{\sigma_z^2} \left(\frac{\phi^k}{\cancel{1 - \phi^2}} \right)}{\cancel{\sigma_z^2} \left(\frac{1}{\cancel{1 - \phi^2}} \right)}$$

$$\rho_k = \phi^k, \quad k = 0, 1, 2, \dots$$

→ If $0 < \phi < 1$ all autocorrelation functions are positive.

→ If $-1 < \phi < 0$ sign of autocorrelation functions shows an alternative pattern beginning with a negative

value

→ The magnitude of the autocorrelation functions decrease exponentially in both cases.

v) Partial AutoCorrelation function (PACF)

we have already discussed some formulas of Partial AutoCorrelation function.
From my notes at page # 72-76

As we know that

$$\rho_k = \phi^k$$

put $k=1$

$$\rho_1 = \phi^1$$

$$\rho_{11} = \rho_1 = \phi$$

put $k=2$

$$\rho_2 = \phi^2$$

ρ is the PACF

$$P_{22} = \frac{\begin{vmatrix} 1 & \rho_1 \\ \rho_1 & \rho_2 \end{vmatrix}}{\begin{vmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{vmatrix}} = \frac{\begin{vmatrix} 1 & \phi \\ \phi & \phi^2 \end{vmatrix}}{\begin{vmatrix} 1 & \phi \\ \phi & 1 \end{vmatrix}}$$

$$P_{22} = \frac{\phi^2 - \phi^2}{1 - \phi^2} = 0$$

Thus

$$\rho_{kk} = 0, \quad k > 1$$

✱. ————— ✱

Stationary Condition of AR(1) process

Consider an AR(1) process

$$Y_t = \phi Y_{t-1} + Z_t$$

We can write the characteristic equation for an AR(P) process

$$x^P - \phi_1 x^{P-1} - \phi_2 x^{P-2} - \dots - \phi_P = 0$$

Thus for an AR(1) process

$$\text{put } P=1$$

$$x^1 - \phi_1 x^0 = 0$$

$$x - \phi_1 = 0$$

$$x = \phi_1$$

$$|x| = |\phi_1| < 1$$

For stationary, The ⁽¹⁰⁰⁾ roots of characteristic equation of AR process must lie inside the unit circle, thus

$$|x| < 1$$

$$|\phi| < 1$$

* . ————— →

AR(2) process

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The autoregressive process of order 2 is defined as,

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$$

It can be written as

$$Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} = Z_t$$

$$Y_t - \phi_1 B Y_t - \phi_2 B^2 Y_t = Z_t$$

$$(1 - \phi_1 B - \phi_2 B^2) Y_t = Z_t$$

$$Y_t = (1 - \phi_1 B - \phi_2 B^2)^{-1} Z_t$$

$$Y_t = \left[1 - (\phi_1 B + \phi_2 B^2) \right]^{-1} Z_t$$

$$\therefore (1-x)^{-1} = 1 + x + x^2 + \dots$$

$$Y_t = [1 - (\phi_1 B + \phi_2 B^2)]^{-1} Z_t \quad (101)$$

$$Y_t = \left[1 + (\phi_1 B + \phi_2 B^2) + (\phi_1 B + \phi_2 B^2)^2 + (\phi_1 B + \phi_2 B^2)^3 + \dots \right] Z_t$$

$$Y_t = \left[1 + \phi_1 B + \phi_2 B^2 + \phi_1^2 B^2 + \phi_2^2 B^4 + 2\phi_1 \phi_2 B^3 + \phi_1^3 B^3 + \phi_2^3 B^6 + 3\phi_1^2 B^4 + \dots \right] Z_t$$

$$Y_t = \left[1 + \phi_1 B + (\phi_2 + \phi_1^2) B^2 + (2\phi_1 \phi_2 + \phi_1^3) B^3 + \dots \right] Z_t$$

$$Y_t = \left[1 + \psi_1 B + \psi_2 B^2 + \psi_3 B^3 + \dots \right] Z_t$$

$$Y_t = \left(\sum_{i=0}^{\infty} \psi_i B^i \right) Z_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$$

i) mean

$$Y_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$$

$$\begin{cases} \psi_1 = \phi_1 \\ \psi_2 = \phi_2 + \phi_1^2 \\ \psi_3 = 2\phi_1 \phi_2 + \phi_1^3 \end{cases}$$

$$E(Y_t) = E \left[\sum_{i=0}^{\infty} \psi_i Z_{t-i} \right]$$

$$= \sum_{i=0}^{\infty} \psi_i E(Z_{t-i})$$

$$= 0$$

ii) Variance

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$$\text{Variance} = E(Y_t^2) - \{E(Y_t)\}^2$$

$$= E(Y_t^2) - 0$$

$$= E(Y_t^2)$$

$$= E\left[\sum_{i=0}^{\infty} \psi_i Z_{t-i}\right]^2$$

$$= E\left[\sum_{i=0}^{\infty} \psi_i^2 Z_{t-i}^2 + \text{C.P term}\right]$$

$$= \sum_{i=0}^{\infty} \psi_i^2 E(Z_{t-i}^2) + 0$$

$$= \sigma_Z^2 \sum_{i=0}^{\infty} \psi_i^2$$

provided that
 $\sum_{i=0}^{\infty} \psi_i^2 < \infty$

iii) Autocovariance

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$$\gamma_k = \text{Cov}(Y_t, Y_{t-k})$$

$$= E(Y_t, Y_{t-k}) - E(Y_t)E(Y_{t-k})$$

$$= E(Y_t Y_{t-k}) - 0$$

$$\gamma_k = E(Y_t Y_{t-k})$$

$$Y_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$$

$$Y_{t-k} = \sum_{j=0}^{\infty} \psi_j Z_{t-k-j}$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \psi_i Z_{t-i} \sum_{j=0}^{\infty} \psi_j Z_{t-k-j} \right]$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j Z_{t-i} Z_{t-k-j} \right]$$

$$\gamma_k = E \left[\sum_{i=0}^{\infty} \sum_{\substack{j=0 \\ j \neq i-k}}^{\infty} \psi_i \psi_j Z_{t-i} Z_{t-k-j} \right]$$

$$+ \sum_{i=0}^{\infty} \sum_{\substack{j=0 \\ j \neq i-k}}^{\infty} \psi_i \psi_j Z_{t-i} Z_{t-k-j}$$

$$\gamma_k = E \left[\sum_{i=k}^{\infty} \psi_i \psi_{i-k} z_{t-i}^2 \right] \quad (104)$$

$$= \sum_{i=k}^{\infty} \psi_i \psi_{i-k} E(z_{t-i}^2) + 0$$

$$\gamma_k = \sigma_z^2 \sum_{i=k}^{\infty} \psi_i \psi_{i-k}$$

$$= \sigma_z^2 (\psi_k \psi_0 + \psi_{k+1} \psi_1 + \psi_{k+2} \psi_2 + \dots)$$

$$\gamma_k = \sigma_z^2 (\psi_0 \psi_k + \psi_1 \psi_{k+1} + \psi_2 \psi_{k+2} + \dots)$$

$$\gamma_k = \sigma_z^2 \sum_{i=0}^{\infty} \psi_i \psi_{k+i}$$

iv) Auto correlation.

$$\begin{aligned} \rho_k &= \frac{\gamma_k}{\gamma_0} \\ &= \frac{\cancel{\sigma_z^2} \sum_{i=k}^{\infty} \psi_i \psi_{i-k}}{\cancel{\sigma_z^2} \sum_{i=0}^{\infty} \psi_i^2} \end{aligned}$$

$$\rho_k = \frac{\sum_{i=k}^{\infty} \psi_i \psi_{i-k}}{\sum_{i=0}^{\infty} \psi_i^2} \quad (6.5), \quad k = 1, 2, 3, \dots$$

v) partial AutoCorrelation Function (PACF)

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + z_t$$

multiply with y_{t-k} and take Expectation

$$E(y_{t-k} y_t) = \phi_1 E(y_{t-k} y_{t-1}) + \phi_2 E(y_{t-k} y_{t-2}) + \dots + \phi_p E(y_{t-k} y_{t-p}) + E(y_{t-k} z_t)$$

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \dots + \phi_p \gamma_{k-p} + 0$$

$\div$ by variance

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \dots + \phi_p \rho_{k-p}, \quad k > 0$$

For AR(2) process

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} \quad \text{--- (1)}, \quad k > 0$$

put $k=1$ in eq (1)

$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1}$$

$$l_1 = \phi_1 + \phi_2 l_1 \quad \text{--- (2)}$$

(106)

put $k=2$ in eq (1)

$$l_2 = \phi_1 l_1 + \phi_2 l_0$$

$$l_2 = \phi_1 l_1 + \phi_2 \quad \text{--- (3)}$$

From eq (2)

$$l_1 - \phi_2 l_1 = \phi_1$$

$$(1 - \phi_2) l_1 = \phi_1$$

$$l_1 = \frac{\phi_1}{1 - \phi_2}$$

Put in eq (3)

$$l_2 = \phi_1 \left[\frac{\phi_1}{1 - \phi_2} \right] + \phi_2$$

$$l_2 = \frac{\phi_1^2}{1 - \phi_2} + \phi_2$$

$$= \frac{\phi_1^2 + \phi_2 (1 - \phi_2)}{1 - \phi_2}$$

$$l_2 = \frac{\phi_1^2 + \phi_2 - \phi_2^2}{1 - \phi_2}$$

$$\ell_2 = \frac{\phi_1^2 + \phi_2 - \phi_2^2}{1 - \phi_2}$$

(107)

AutoCorrelation کو
Final کیا ہے۔ یہ فردی میں
AC کیونکہ ہم نے دیکھا
Final کر رہا ہے۔ دیباچہ میں صرف
 ℓ_1 کی جگہ طے ہے۔

ℓ_k for $k \geq 3$ is calculated recursively

Now, Partial AutoCorrelation (PACF)

As we know that

$$\phi_{11} = \ell_1 = \frac{\phi_1}{1 - \phi_2}$$

$$\phi_{22} = \frac{\begin{vmatrix} 1 & \ell_1 \\ \ell_1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & \ell_1 & \ell_2 \\ \ell_1 & 1 & \ell_1 \\ \ell_2 & \ell_1 & 1 \end{vmatrix}} = \frac{\ell_2 - \ell_1^2}{1 - \ell_1^2}$$

$$= \frac{\left(\frac{\phi_1^2 + \phi_2 - \phi_2^2}{1 - \phi_2} \right) - \left(\frac{\phi_1}{1 - \phi_2} \right)^2}{1 - \left(\frac{\phi_1}{1 - \phi_2} \right)^2}$$

$$\phi_{22} = \frac{\phi_2 \left[(1 - \phi_2)^2 - \phi_1^2 \right]}{(1 - \phi_2)^2 - \phi_1^2} = \phi_2 \quad (108)$$

$$\phi_{33} = \frac{\begin{vmatrix} 1 & e_1 & e_1 \\ e_1 & 1 & e_2 \\ e_2 & e_1 & e_3 \end{vmatrix}}{\begin{vmatrix} 1 & e_1 & e_2 \\ e_1 & 1 & e_1 \\ e_2 & e_1 & 1 \end{vmatrix}}$$

put the values of eq (2) and (3)

$$\phi_{33} = \frac{\begin{vmatrix} 1 & e_1 & \phi_1 + \phi_2 e_1 \\ e_1 & 1 & \phi_1 e_1 + \phi_2 \\ e_2 & e_1 & \phi_1 e_2 + \phi_2 e_1 \end{vmatrix}}{\begin{vmatrix} 1 & e_1 & e_2 \\ e_1 & 1 & e_1 \\ e_2 & e_1 & 1 \end{vmatrix}} = 0$$

$$\begin{vmatrix} 1 & e_1 & e_2 \\ e_1 & 1 & e_1 \\ e_2 & e_1 & 1 \end{vmatrix}$$

$$\phi_{kk} = 0 \quad \text{for } k \geq 3$$

Because it is AR(2), order is 2, $p=2$

Stationary Condition of AR(2) process

Consider an AR(2) process

$$Y_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

We can write the characteristic eq

$$x^p - \phi_1 x^{p-1} - \phi_2 x^{p-2} - \dots - \phi_p = 0$$

put $p=2$

$$x^2 - \phi_1 x - \phi_2 = 0$$

$$x^2 - \phi_1 x - \phi_2 = 0$$

The roots of above equation can be obtained using quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2}$$

We have two roots.

$$x_1 = \frac{-\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2}, \quad x_2 = \frac{-\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2} \quad (110)$$

For stationary

$$|x_1| < 1 \quad \text{and} \quad |x_2| < 1$$

$$\Rightarrow |x_1 x_2| < 1$$

First take $|x_1| < 1$

$$\left| \frac{-\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| < 1$$

$$|-\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}| < 2$$

$$-2 < \phi_1 + \sqrt{\phi_1^2 + 4\phi_2} < 2$$

$$-2 - \phi_1 < \sqrt{\phi_1^2 + 4\phi_2} < 2 - \phi_1$$

Now, we consider the lower bound.

$$-2 - \phi_1 < \sqrt{\phi_1^2 + 4\phi_2}$$

$$-(2 + \phi_1) < \sqrt{\phi_1^2 + 4\phi_2}$$

S.B.S

(iii)

$$(2 + \phi_1)^2 > \phi_1^2 + 4\phi_2$$

$$4 + \cancel{\phi_1^2} + 4\phi_1 > \cancel{\phi_1^2} + 4\phi_2$$

$$4 > \cancel{\phi_1^2} + 4\phi_2 - \cancel{\phi_1^2} - 4\phi_1$$

$$4 > 4\phi_2 - 4\phi_1$$

$$4 > 4(\phi_2 - \phi_1)$$

$$\boxed{\phi_2 - \phi_1 < 1}$$

Now, we take upper bound

$$\sqrt{\phi_1^2 + 4\phi_2} < 2 - \phi_1$$

S.B.S

$$\phi_1^2 + 4\phi_2 < (2 - \phi_1)^2$$

$$\phi_1^2 + 4\phi_2 < 4 + \phi_1^2 - 4\phi_1$$

$$\cancel{\phi_1^2} + 4\phi_2 - \cancel{\phi_1^2} + 4\phi_1 < 4$$

$$4(\phi_2 + \phi_1) < 4$$

$$\boxed{\phi_2 + \phi_1 < 1}$$

Secondly, we take (ii) $|x_1 x_2| < 1$

$$|x_1 x_2| < 1$$

$$\left| \frac{-\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2} \cdot \frac{-\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| < 1$$

$$\left| \frac{\phi_1^2 - (\phi_1^2 + 4\phi_2)}{4} \right| < 1$$

$$\left| \frac{\cancel{\phi_1^2} - \cancel{\phi_1^2} - 4\phi_2}{4} \right| < 1$$

$$\left| \frac{-\cancel{4}\phi_2}{\cancel{4}} \right| < 1$$

$$|-\phi_2| < 1$$

$$|\phi_2| < 1$$

$$\boxed{-1 < \phi_2 < 1}$$

Thus an AR(2) process will be stationary if it satisfy the following three condition.

$$\phi_2 - \phi_1 < 1$$

$$\phi_2 + \phi_1 < 1$$

$$-1 < \phi_2 < 1$$

اگر ان تینوں میں سے
Condition ایک بھی
satisfy نہیں ہوگی
تو stationary نہیں ہوگا۔

